gpeyt1-48 · Uplift capacity in clay

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A transmission tower leg is founded on an 18 in. diameter drilled concrete pier, 40 ft long, in uniform stiff clay (su = 1,100 psf, adhesion factor α = 0.7). The water table is deep (below the pier). Concrete γc = 150 pcf. Apply FS = 3.0 to side resistance only; pier self-weight is counted at full value (unfactored). Ignore any belling and end suction.

Most nearly, the allowable uplift capacity is:

A. 52 kips
B. 59 kips
C. 48 kips
D. 55 kips

Given equations / data

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Tall = α su π D L / FS + Wpier
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Worked solution

1. Qs = 0.7×1,100×π×1.50×40 = 145,142 lb
2. W = π/4×1.50²×40×150 = 10,603 lb
3. Tall = 145,142/3.0 + 10,603 = 58,983 lb ≈ 59 kips

Final answer

59 kips

Why the other values are wrong

- **52 kips** — applied FS to pier weight too
- **48 kips** — ignored pier weight
- **55 kips** — used buoyant concrete weight though WT is deep

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) α-method side resistance; uplift Rn = Rs + W (Eq. 7-41 group form)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=jnAwn45noZw (no creator values copied).

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